

13.2 Evaluating Algebraic Expressions Using Substitution

Lesson Introduction

 Benchmark	MA.6.AR.1.3 Evaluate algebraic expressions using substitution and order of operations.		 Learning Target	I can evaluate algebraic expressions using substitution.	
 Benchmark Coherence	Building On		Working Toward		
	MA.5.AR.2 Demonstrate an understanding of equality, the order of operations, and equivalent numerical expressions.		MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers including grouping symbols, whole-number exponents, and absolute value.		
 Essential Questions	- How do you evaluate algebraic expressions using substitution? - How does order of operations influence the outcome when simplifying an expression?				
 Lesson Narrative	Lesson 1 positioned students to evaluate algebraic expressions using substitution and order of operations. This lesson is designed to consolidate student understanding by analyzing samples of student work for errors. Students engage in error analysis to solidify their own conceptual understanding before applying to practice.				
 Component Summary	13.2.1 Warm-Up (~5 min.)	Create an exponential expression with the largest value (<i>SE p. 270</i>)	13.2.3 Your Turn! (15 min.)	Practice evaluating algebraic expressions (<i>SE p. 272</i>)	
	13.2.2 Collaboration (20 min.)	Analyze errors and evaluate algebraic expressions (<i>SE p. 271</i>)	13.2.4 Wrap-Up (~5 min.)	Self-assessment of understanding of evaluating algebraic expressions (<i>SE p. 273</i>)	
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> Evaluate		(Optional) Calculators (Optional) colored pencils		If differentiating by ability level, consider planning for how to collect informal data throughout the lesson to inform which expressions to assign to students. It may be helpful to use student work and explanations during 13.2.2 Collaboration as formative data.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may substitute the variable as a digit instead of multiplying it by the coefficient. For example, in $5x$, where $x = 4$, students may write 54 instead of $5(4)$. - Students may evaluate exponents as the base multiplied by the exponent instead of the base multiplied by itself the exponent number of times. - Students may struggle remembering the rules for operations with integers.	- Encourage students to substitute one variable at a time and to use parentheses and/or colored pencils. This may help students keep track of their work and evaluate expressions more accurately. - Encourage students to evaluate one operation at a time, underline the calculation at each step, and show all work.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share Use Think-Pair-Share in 13.2.2 Collaboration to give all students enough time to think about a prompt and form a response before they are expected to try to verbalize their thinking. Students will have an opportunity to share their thinking in a low-stakes way with one partner so that when they share with the class, they can feel calm and confident as well as say something meaningful that might advance everyone’s understanding. Additionally, the teacher can listen in so that they can purposefully select students to share with the class.	MA.K12.MTR.4.1: Discussion & Reflection During 13.2.2 Collaboration, students will be engaging in discourse to find and correct an error representing a common misconception that students make when evaluating algebraic expressions. This standard emphasizes students reflecting on their own work and the work of others to establish a culture in which mistakes are viewed as an opportunity for learning.
 Differentiate Instruction	Consider having students preview this lesson by watching the On-Ramp video on evaluating expressions or have the video available to watch for students to opt-in throughout this lesson, particularly during 13.2.3 Your Turn!	
Lesson Notes:		

13.2.1 Warm-Up: Order Explorer

Component Overview: Students work independently to use digits one through five to create an exponential expression with the largest possible value. After students have completed questions 1-3, bring the class together to review the expressions students created and their strategies for determining the largest value.



13.2.1 Warm-Up: Order Explorer

- Place a digit in each box to create an expression with the largest possible value. Use any of the digits 1 to 5, once each at most.

$$\boxed{4}^{\boxed{5}} + \boxed{2} \times \boxed{3}$$

- What is the value of your expression?
1,030
- On a scale of 1 to 10, how confident are you that your expression is the largest possible value?
Answers will vary.
- Explain your strategy for choosing and placing the numbers as you did.
I calculated using both 4^5 and 5^4 and once I realized that 4^5 was larger I then multiplied by the next largest numbers.



The goal of 13.2.1 Warm-Up is to be an open-ended activity for the student. The teacher should resist guiding students with how to approach the task. Instead, the questions below can be used to facilitate a deeper conversation around the type of thinking and reasoning students used to determine where to place each digit within the expression.



- Did you try multiple expressions to determine which would be the largest?
- Which digit did you choose not to use? Why?
- Where did you place the largest digit, five?
- Why did you choose to place it there?
- How did you decide where to place the remaining digits to create an expression with the largest possible value?

13.2.2 Collaboration: Evaluating Algebraic Expressions by Substituting Integers

Component Overview: Students work with a partner to find the error in student work provided. Engage students in a Think-Pair-Share routine before having them complete the written response for 1b. Consider implementing a whole-class discussion before students move on to question 2 to solidify understanding and address any questions students may have after completing question 1. Students then work independently to evaluate Expressions 2a-2c before checking their work with a partner.



13.2.2 Collaboration: Evaluating Algebraic Expressions by Substituting Integers

1. Brae evaluated the expression $7b + 12 - (5 - 3)^2$ for $b = -2$. Her work is shown.
 - a. Evaluate her work and determine where she made mistakes.

$$\begin{aligned}
 &7b + 12 - (5 - 3)^2 \\
 &7 - 2 + 12 - (5 - 3)^2 \\
 &7 - 2 + 12 - (-2)^2 \\
 &7 - 2 + 12 + 4 \\
 &5 + 12 + 4 \\
 &17 + 4 \\
 &21
 \end{aligned}$$

- b. Help Brae understand and learn from her mistakes. Write a brief note to Brae explaining what her mistakes were and how she could learn from the mistakes.

Means

I

Start

To

Acquire

Knowledge

Experience

Skills

When Brae substituted -2 in for $7b$ she forgot to put the -2 in parentheses. This made it look like $7 - 2$ instead of 7 times -2 . Brae should be careful and use parentheses when substituting a value in.

When Brae found $5 - 3$ she wrote -2 instead of 2 .

- c. Read the note your partner wrote and offer suggestions on ways to improve their note.



Think-Pair-Share

Read the directions for Part b to the class, then have students engage in a Think-Pair-Share with a partner. Encourage students to use the following format to answer the question completely:

1. Name the error(s).
2. Explain the error. Why do you think Brae made this mistake?
3. What does Brae need to remember to prevent making this mistake in the future?

Providing a structure for this discussion and the following response will help students think through the misconceptions presented in this question.

13.2.2 Collaboration: Evaluating Algebraic Expressions by Substituting Integers (continued)

- d. Using the word bank below, complete the three reminders that could help Brae.

inside outside parentheses

- When substituting values into algebraic expressions, it is helpful to put the substituted value(s) in *parentheses*.
 - When evaluating expressions with exponents, if the negative is *outside* the parentheses, the negative indicates subtraction, or the opposite of the value found by raising the value inside to a power.
 - If the negative is *inside* the parentheses, it indicates that the number is a negative number being raised to a power.
- e. Verify your answers with your partner and, if necessary, resolve any differences.



Consider a Read Aloud with a small group for students who struggle with reading comprehension. Alternatively, nominate reading captains for each small group that read for their respective groups.

13.2.2 Collaboration: Evaluating Algebraic Expressions by Substituting Integers (continued)

2. Complete the following problems.

- a. Evaluate the expression $-8m - 4q^2 + \frac{s}{3}$ where $m = 4$, $q = -3$, and $s = -24$.

$$\begin{aligned} & -8(4) - 4(-3)^2 + \frac{-24}{3} \\ & -32 - 4(9) - 8 \\ & -32 - 36 - 8 = -76 \end{aligned}$$

- b. Evaluate the expression $3s - 4q - m^2$ where $m = 4$, $q = -3$, and $s = -24$.

$$\begin{aligned} & 3(-24) - 4(-3) - (4)^2 \\ & -72 + 12 - 16 \\ & -76 \end{aligned}$$

- c. Evaluate the expression $-m^3 - 3q + (s - q)$ where $m = 4$, $q = -3$, and $s = -24$.

$$\begin{aligned} & -4^3 - 3(-3) + (-24 - (-3)) \\ & -64 + 9 + (-24 + 3) \\ & -64 + 9 + (-21) \\ & -64 - 21 + 9 \\ & -85 + 9 = -76 \end{aligned}$$

- d. All of the expressions should have the same result. Check your work with your partner.



- How can you break this question down into steps to make it easier to solve?
- What happens when you substitute each variable for a number?
- How can you use grouping symbols like parentheses to keep track of substituting variables for numbers?

13.2.3 Your Turn!

Component Overview: Students work independently to evaluate expressions by substituting variables.



13.2.3 Your Turn!

1. What is the value of $5(6x + 7) - 3y^2$ if $x = -1$ and $y = 5$?

$$\begin{aligned} &5(6(-1) + 7) - 3(5)^2 \\ &5(-6 + 7) - 3(25) \\ &5(1) - 75 \\ &5 - 75 = -70 \end{aligned}$$

2. Evaluate the expression $3d^4 - \frac{w}{3}$ where $d = 4$ and $w = -75$.

$$\begin{aligned} &3(4)^4 - \frac{(-75)}{3} \\ &3(256) + \frac{75}{3} \\ &768 + 25 = 793 \end{aligned}$$

$$\begin{aligned} &4^4 = 4 \cdot 4 \cdot 4 \cdot 4 \\ &\quad \swarrow \quad \searrow \\ &16 \quad 16 \\ &\quad \swarrow \quad \searrow \\ &256 \\ &256 \cdot 3 = 768 \end{aligned}$$

3. Find the value of $-2(b + 4) + 3a - (7 - a)^5$ if $a = -3$ and $b = 5$.

$$\begin{aligned} &-2(5 + 4) + 3(-3) - (7 - (-3))^5 \\ &-2(9) + (-9) - (7 + 3)^5 \\ &-2(9) + (-9) - (10)^5 \\ &-18 + (-9) - 100,000 \\ &= -100,027 \end{aligned}$$

$$\begin{aligned} &10 \cdot 10 \cdot 10 \cdot 10 \cdot 10 \\ &\quad \swarrow \quad \searrow \\ &100 \cdot 100 \cdot 10 \\ &10000 \cdot 10 = 100,000 \end{aligned}$$

4. Evaluate the expression $c^3 - 5k + (h - k)$ where $c = -9$, $h = 29$, and $k = -17$.

$$\begin{aligned} &(-9)^3 - 5(-17) + (29 - (-17)) \\ &(-9)^3 - 5(-17) + (29 + 17) \\ &-729 + 85 + 46 \\ &-729 + 131 \\ &= -598 \end{aligned}$$

$$\begin{aligned} &-9 \cdot -9 \cdot -9 \\ &81 \cdot -9 = -729 \\ &729 - 131 = 598 \end{aligned}$$



Consider strategies for differentiating this activity based on your students and classroom.

- Provide On-Ramp videos for students to opt-in for additional support before engaging in the practice.
- Conduct an open small group in which students can come and go as they need assistance.
- Allow student choice. Students can choose any two questions to solve before finding a partner who did the other two questions to then explain (not copy) their respective answers.



Each of the questions in 13.2.3 Your Turn! highlight some common misconceptions and errors that students may make.

Question 1 – Look for students who multiply -3 and 5 before squaring.

Question 2 – Look for students who will drop the negative from 75 .

Question 3 – Look for students who distribute the subtraction sign in front of $(7 - a)$ to the 7 before finding the value inside the parentheses.

Consider having students do the questions, and then check their work and answers with a partner and to work through any differences before completing question 5.

13.2.3 Your Turn! (continued)

5. Evaluate each expression.

a. $8 \div p - 16 \div (-5 + 7)$ where $p = 4$

$$8 \div 4 - 16 \div (2)$$

$$2 - 8 = -6$$

b. $13 + \frac{6}{y} - y^3$ if $y = -3$

$$13 + \frac{6}{-3} - (-3)^3$$

$$13 + (-2) - (-27)$$

$$13 - 2 + 27$$

$$11 + 27 = 38$$

c. $4g - 12 + (h - 7) \cdot 3^2$ where $g = -7$ and $h = 6$

$$4(-7) - 12 + (6 - 7) \cdot 9$$

$$-28 - 12 + (-9)$$

$$-28 - 12 - 9$$

$$= -49$$

d. $w \div t \cdot H \div y^2$ where $H = -3$, $t = 7$, $w = 84$, and $y = -2$

$$84 \div 7 \cdot (-3) \div (-2)^2$$

$$12 \cdot (-3) \div 4$$

$$-36 \div 4 = -9$$



Consider providing a four-function calculator to students who struggle with computation. A four-function calculator is not programmed to perform the order of operations so the student will still have to determine which operation to perform first. Allowing the use of a calculator may give students the chance to master the learning target without getting caught up in complex calculations. Have students write and say the calculations they enter in the calculator to reinforce their computational skills.

13.2.4 Wrap-Up: Reflection

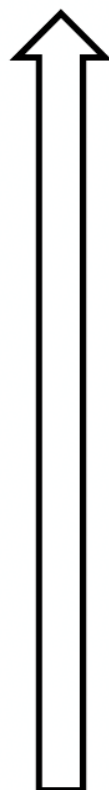
Component Overview: Students complete a self-assessment of their understanding of solving algebraic expressions. Consider using this assessment to inform small group remediation targets.



13.2.4 Wrap-Up: Reflection

Color in the arrow up to the statement that best describes your current understanding.

Answers will vary.



I'm so confident, I can explain evaluating algebraic expressions using substitution to someone else!

I can get to the right answer, but I don't understand well enough to explain it yet.

I understand some of this, but I don't understand all of it yet. I checked what I do not understand yet.

- ☐ Working with negative numbers
- ☐ Finding values raised to an exponent
- ☐ Remembering to multiply and divide from left to right
- ☐ Remembering to use parentheses when I substitute a value

I tried hard, but I am finding this challenging. I checked what I do not understand yet.

- ☐ Working with negative numbers
- ☐ Finding values raised to an exponent
- ☐ Remembering to multiply and divide from left to right
- ☐ Remembering to use parentheses when I substitute in a value



Students could describe their understanding of each learning goal verbally by coloring in the scale or by using a coding system (i.e. 0=not at all, 1=partial, 2=full).